

1.8 Integration

Pure Mathematics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Integration reverses differentiation up to a constant C . For a definite integral use antiderivative limits; signed area below x -axis is negative. A substitution reverses the chain rule and usually changes integration limits in a definite integral.

Worked example: follow the reasoning

$\int (3x^2 - 2) dx = x^3 - 2x + C$; from 0 to 2 the signed integral is $8 - 4 = 4$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Evaluate $\int$ from 1 to 3 of $2x \, dx$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Antiderivative x^2 [1]; x^2 evaluated 1 to 3 [1] $=9-1=8$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.